

APPASA Strategy Document

Stage 1 – Ask: APPASA Business Understanding & Stakeholder Alignment



APPASA: Ask Stage

Introduction

I will use this **APPASA strategy document** to record my decisions and reflections as I work through **Stage 1: Ask** of this project. This document will serve as a guide, helping me consider my responses and reflections throughout this stage of the data analysis process.

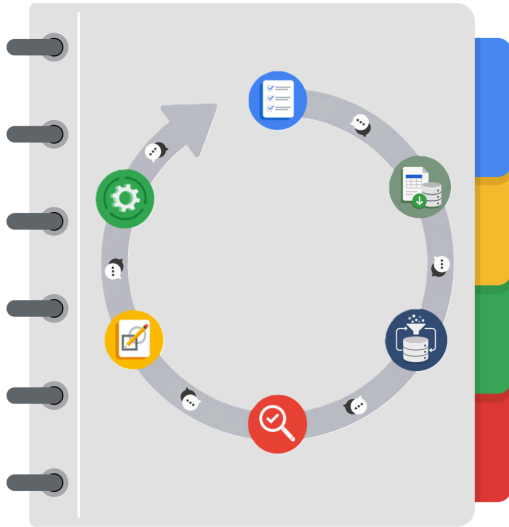
Portfolio Project Recap

Many of the goals I accomplished throughout the individual course portfolio projects are integrated into this end-to-end data analysis capstone project. These include:

- Creating a clear and structured project proposal.
- Demonstrating my understanding of SQL and relational database concepts.
- Using SQL to ingest, explore, clean, transform, validate, and organize data through structured queries and database operations.
- Organizing and analyzing a large-scale dataset to uncover meaningful insights and communicate the underlying business story.
- Developing a series of SQL scripts for ELT, feature engineering, data validation, and analytical querying, alongside a DuckDB-powered Jupyter Notebook documenting the complete APPASA methodology and reproducing the analytical workflow through SQL queries and visualizations.
- Computing descriptive statistics, performing exploratory and business-focused analyses, and creating analytical visualizations to support data interpretation and executive decision-making.
- Communicating findings through an executive summary, an interactive Power BI report, strategic business recommendations, and implementation-focused action plans for stakeholders.
- Translating analytical findings into evidence-based business recommendations, implementation strategies, success metrics, and future analytical opportunities through the final Act stage of the APPASA framework.

This capstone brings together the knowledge and technical skills I developed throughout the program, demonstrating my ability to execute a complete end-to-end data analysis project, from business problem definition to actionable business recommendations.

THE APPASA WORKFLOW



I will demonstrate to the company's marketing analytics team how I would apply the APPASA workflow to the upcoming Cyclistic Bike Share project. For each question presented in this APPASA strategy document, I will provide a structured and thoughtful response aligned with the corresponding stage of the APPASA framework: Ask, Prepare, Process, Analyze, Share and Act. This approach ensures clarity in my methodology, highlights my strategic planning skills, and illustrates a well-organized path from problem understanding to actionable insights.

Project Tasks

The following questions have been identified as essential to the Member vs Casual Rider Behavior Analysis Project. These questions are primarily situated within the Ask stage of the APPASA workflow, focusing on understanding the business problem, identifying key stakeholders, defining project objectives, and establishing project expectations. I will address each question by drawing on relevant materials from the Cyclistic case study, the APPASA framework, project documentation, and best practices for business understanding and stakeholder alignment.





Data Project Questions & Considerations

Business Understanding

- Who are the key stakeholders involved in this project?

The key stakeholders involved in this project are:

- **Lily Moreno (Director of Marketing):** As the project sponsor and my manager, Lily Moreno is responsible for developing marketing campaigns and strategic initiatives to promote the Cyclistic bike-share program. She has tasked the marketing analytics team with understanding how annual members and casual riders use Cyclistic bikes differently to support the development of targeted membership conversion strategies.
- **Cyclistic Marketing Analytics Team:** As a junior data analyst, I am part of the marketing analytics team responsible for collecting, analyzing, and communicating data-driven insights that support Cyclistic's marketing objectives. The team's findings will guide the development of evidence-based marketing recommendations.
- **Cyclistic Executive Team:** The executive team will review the analytical findings, visualizations, and recommendations before approving any proposed marketing strategy. They require clear, accurate, and well-supported insights to make informed business decisions regarding future membership growth initiatives.

These stakeholders play a critical role throughout the project, from defining the business objectives to evaluating the final recommendations aimed at increasing the number of annual memberships.

- What business problem is Cyclistic trying to solve?

Cyclistic is trying to address the challenge of increasing the number of annual memberships to support the company's long-term growth and profitability. While the company's flexible pricing structure, including single-ride passes, full-day passes, and annual memberships, has successfully attracted a broad customer base, Cyclistic's finance analysts have concluded that annual members are significantly more profitable than casual riders.

Rather than focusing its marketing efforts on acquiring entirely new customers, Cyclistic aims to convert existing casual riders into annual members. Since casual riders are already familiar with the bike-share program and actively use its services, they represent a valuable opportunity for sustainable membership growth.

To achieve this objective, Cyclistic must first understand how annual members and casual riders differ in their riding behavior. These insights will enable the marketing team to develop targeted, data-driven marketing strategies that encourage casual riders to purchase annual memberships while supporting informed decision-making by the executive team.



- What are the primary business objectives of this project?

The primary business objective of this project is to support Cyclistic's long-term growth by increasing the number of annual memberships. To achieve this, the project aims to identify and understand the behavioral differences between annual members and casual riders through the analysis of Cyclistic's historical bike trip data.

By uncovering meaningful patterns in rider behavior, the project seeks to generate actionable insights that will help the marketing team develop targeted strategies for converting casual riders into annual members. These findings will also provide the executive team with reliable, data-driven evidence to evaluate and approve future marketing initiatives.

Ultimately, the project aims to transform historical trip data into strategic business insights that support informed decision-making, improve marketing effectiveness, and contribute to sustainable membership growth.

- What business task has been assigned to me?

As a junior data analyst on the Cyclistic marketing analytics team, I have been assigned to investigate how annual members and casual riders use Cyclistic bikes differently. This analysis focuses on the first of the three business questions identified by the marketing team and serves as the foundation for developing future membership conversion strategies.

The business question assigned to me is:

1. How do annual members and casual riders use Cyclistic bikes differently?

My responsibility is to analyze Cyclistic's historical bike trip data, identify meaningful behavioral differences between annual members and casual riders, and communicate these findings through clear, data-driven insights and professional visualizations. Based on the analysis, I will provide actionable recommendations that support the marketing team's objective of converting casual riders into annual members and assist the executive team in making informed business decisions.

- What business question(s) should this project answer?

This project is guided by three key business questions identified by the Cyclistic marketing team:

1. How do annual members and casual riders use Cyclistic bikes differently?

2. Why would casual riders purchase Cyclistic annual memberships?

3. How can Cyclistic use digital media to encourage casual riders to become annual members?

As the assigned data analyst, my primary responsibility is to answer the first question by analyzing Cyclistic's historical bike trip data to identify behavioral differences between annual members and casual riders. The

insights generated from this analysis will serve as the foundation for addressing the remaining business questions and developing targeted marketing strategies aimed at increasing annual memberships.

- Why is this project important to Cyclistic?

This project is important to Cyclistic because increasing the number of annual memberships is a key driver of the company's long-term growth and profitability. Although Cyclistic's flexible pricing model has successfully attracted both casual riders and annual members, the company's finance analysts have determined that annual members generate greater long-term value than casual riders.

By understanding how annual members and casual riders differ in their riding behavior, Cyclistic can develop more effective and targeted marketing strategies that encourage casual riders to purchase annual memberships. This approach allows the company to focus on converting existing customers who are already familiar with the service, rather than relying solely on acquiring new customers.

The insights generated through this project will support data-driven decision-making, strengthen future marketing initiatives, and contribute to sustainable business growth by increasing customer retention and membership adoption.

- How can the insights generated from this project support business decisions?

The insights generated from this project will enable Cyclistic to make informed, data-driven business decisions by providing a deeper understanding of how annual members and casual riders differ in their riding behavior. Identifying these behavioral patterns will help the marketing team design targeted campaigns and promotional strategies that address the specific needs and preferences of casual riders.

The findings will also support the executive team in evaluating and approving marketing initiatives by providing clear evidence to justify strategic decisions. By understanding when, where, and how different customer groups use Cyclistic bikes, the company can allocate marketing resources more effectively, improve membership conversion efforts, and maximize the long-term profitability of its customer base.

Ultimately, these insights will help Cyclistic develop marketing strategies that encourage more casual riders to become annual members, supporting sustainable business growth and increasing the overall value generated from its existing customers.

- What deliverables will be produced throughout this project?

Throughout this project, I will produce a comprehensive set of technical and business-focused deliverables that document the complete analytics workflow and communicate the project's findings effectively. These deliverables include:

- **A project proposal** outlining the project's objectives, scope, deliverables, estimated timeline, and planned analytical workflow.



- **APPASA strategy documents** for each stage of the project, documenting key decisions, reflections, and planning throughout the analytics workflow.
- **A collection of SQL scripts** covering data ingestion, preprocessing, feature engineering, data validation, and analytical querying.
- **A DuckDB-powered Jupyter Notebook** documenting the complete APPASA methodology, analytical workflow, SQL queries, visualizations, and business insights.
- **An interactive Power BI report** presenting rider behavior, ride location analysis, and strategic business recommendations.
- **A Power BI report PDF** for offline viewing and presentation purposes.
- **A report walkthrough video** demonstrating the functionality and insights of the Power BI report.
- **An executive summary** highlighting the project's objectives, methodology, key findings, and recommendations.
- **A GitHub repository** containing the project documentation, source code, supporting assets, and final deliverables.

These deliverables collectively demonstrate the complete end-to-end analytics process, from business understanding and project planning to data preparation, analysis, visualization, and strategic decision-making.

Stakeholder Alignment

- Who is the intended audience for this project?

The intended audience for this project includes the Cyclistic marketing analytics team, Lily Moreno, the Director of Marketing, and the Cyclistic executive team. Each audience has a distinct role in the decision-making process and will use the project's findings in different ways.

The marketing analytics team will use the analysis to better understand rider behavior and develop evidence-based marketing strategies. Lily Moreno will use the insights to design initiatives aimed at converting casual riders into annual members and to communicate the value of these strategies to senior leadership. The executive team will review the analytical findings, visualizations, and business recommendations to evaluate and approve future marketing campaigns.

Because the audience consists of both technical and non-technical stakeholders, the findings must be communicated through clear explanations, professional data visualizations, and actionable business recommendations that support informed decision-making.



- What are the expectations of each stakeholder?

Each stakeholder has specific expectations based on their role in the project and how they will use the analytical findings.

- **Lily Moreno (Director of Marketing):** Expects a comprehensive analysis of rider behavior that explains how annual members and casual riders differ. She expects actionable, data-driven recommendations that can be translated into effective marketing strategies for increasing annual memberships.
- **Cyclistic Marketing Analytics Team:** Expects accurate, reliable, and well-documented analyses that provide meaningful insights into customer behavior. The team relies on these findings to support strategic planning, campaign development, and future marketing initiatives.
- **Cyclistic Executive Team:** Expects clear, concise, and evidence-based findings supported by professional data visualizations. Before approving any proposed marketing program, the executive team requires confidence that the recommendations are backed by robust analysis and aligned with Cyclistic's long-term business objectives.

By meeting these expectations, the project will provide stakeholders with the information needed to make informed decisions that support membership growth and improve the effectiveness of Cyclistic's marketing strategy.

- What decisions could stakeholders make based on this analysis?

The findings from this project will enable stakeholders to make informed business decisions aimed at increasing the number of annual memberships and improving the effectiveness of Cyclistic's marketing strategy. By understanding how annual members and casual riders differ in their riding behavior, stakeholders can identify opportunities to better engage casual riders and encourage long-term membership adoption.

Based on the analysis, stakeholders may decide to develop targeted marketing campaigns for specific rider segments, introduce promotional offers or membership incentives, optimize digital marketing strategies, and allocate marketing resources toward locations, seasons, or time periods with the greatest conversion potential. The insights may also influence future pricing strategies, customer engagement initiatives, and partnership opportunities designed to attract and retain annual members.

Ultimately, these decisions will help Cyclistic strengthen customer retention, maximize the value of its existing customer base, and support sustainable business growth through data-driven marketing initiatives.



Project Planning

- What analytical approach will I follow to complete this project?

To complete this project, I will follow the APPASA data analytics methodology, which consists of six sequential stages: Ask, Prepare, Process, Analyze, Share, and Act. This structured approach provides a systematic framework for understanding the business problem, preparing and processing the data, performing meaningful analyses, communicating insights, and developing actionable business recommendations.

Throughout the project, I will use SQL to perform data ingestion, preprocessing, feature engineering, data validation, and analytical querying. A DuckDB-powered Jupyter Notebook will document the complete analytical workflow, reproduce the SQL analyses, and present supporting visualizations and interpretations. Finally, I will develop an interactive Power BI report to communicate the project's findings through business-focused dashboards, data storytelling, and strategic recommendations.

By following this structured methodology, I aim to ensure that every stage of the project is well documented, reproducible, and aligned with Cyclistic's business objectives while delivering reliable, data-driven insights that support informed decision-making.

- What tools and technologies are expected to be used throughout this project?

This project will leverage a combination of database, analytics, visualization, and documentation tools to support the complete end-to-end data analytics workflow. Each tool has been selected based on its strengths and role within the APPASA methodology.

The primary tools and technologies expected to be used throughout this project include:

- **MySQL:** For data ingestion, preprocessing, transformation, feature engineering, data validation, and analytical querying.
- **DuckDB:** For executing SQL queries within a Jupyter Notebook and documenting the complete analytical workflow.
- **Jupyter Notebook:** For documenting the APPASA methodology, SQL analyses, visualizations, interpretations, and business insights.
- **Python:** For creating analytical visualizations and supporting the notebook environment where required.
- **Power BI:** For developing an interactive business intelligence report that communicates key findings and strategic recommendations.
- **GitHub:** For hosting the project repository, documentation, SQL scripts, notebooks, Power BI assets, supporting project files, and managing project versions.

Together, these tools provide a robust and reproducible analytics workflow, enabling efficient data processing, comprehensive analysis, effective visualization, and clear communication of business insights.

- What defines the successful completion of this project?

The successful completion of this project will be defined by its ability to answer the assigned business question and provide actionable, data-driven recommendations that support Cyclistic's objective of increasing annual memberships. Success will be measured not only by the quality of the analysis, but also by the clarity, reliability, and practical value of the insights delivered to stakeholders.

Specifically, the project will be considered successful if it:

- Clearly identifies and explains the behavioral differences between annual members and casual riders.
- Produces accurate, well-documented, and reproducible analyses supported by SQL queries and appropriate data validation.
- Communicates findings effectively through professional visualizations, an interactive Power BI report, and an executive summary.
- Provides strategic business recommendations that can help the marketing team design initiatives to convert casual riders into annual members.
- Demonstrates a complete end-to-end analytics workflow by following the APPASA methodology, from business understanding through actionable recommendations.

By achieving these objectives, the project will deliver meaningful business value while demonstrating a structured, professional, and reproducible approach to solving a real-world data analytics problem.

Reflection

- What assumptions or expectations do I currently have before beginning the project?

Before beginning the analysis, I expect that annual members and casual riders will exhibit distinct riding patterns and usage behaviors. Based on the project scenario, I anticipate that annual members are more likely to use Cyclistic bikes for regular commuting and routine travel, whereas casual riders are more likely to use the service for leisure, recreation, or occasional trips.

I also expect to observe differences in ride duration, ride frequency, preferred ride times, seasonal usage, bike type preferences, and popular start and end stations between the two rider groups. These behavioral differences may provide valuable opportunities for developing targeted marketing strategies that encourage casual riders to purchase annual memberships.



While these expectations provide an initial direction for the analysis, they will be validated or refuted through a comprehensive examination of Cyclistic's historical bike trip data. Any conclusions drawn from this project will be based solely on evidence obtained through data analysis rather than prior assumptions.